

Faculty: BSC(Hons) Computer System Engineering

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MODULE CODE: CET138

MODULE TITLE: Full Stack Development

MODULE LEADER: David Grey

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ASSIGNMENT: 1

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CET138 Full Stack Development

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Live Demo link of Bootstrap: <https://rahulchaudhary12-debug.github.io/Demos/demo-bootstrap.html>

Bootstrap Framework

What is Bootstrap ?

The Bootstrap is one of the world's most popular open-source CSS frameworks for creating responsive, mobile-first websites. The CodeCraft Bootstrap demo (demo-bootstrap.html) is a complete repeat of the CodeCraft web training site as a fully responsive HTML page, built using only Bootstrap utility classes and components and without writing any custom CSS.

```
<!-- Bootstrap CSS via CDN - added in <head> -->
<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" />

<!-- Bootstrap JS bundle (includes Popper) - before </body> -->
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></scr
ipt>
```

Responsive Navbar

```
<nav class="navbar navbar-expand-lg navbar-dark bg-black border-bottom border-
secondary">
  <div class="container">
    <a class="navbar-brand fw-bold text-success" href="#">
      Code<span class="text-white">Craft</span>
    </a>
    <!-- Hamburger button - shown on mobile only -->
    <button class="navbar-toggler" type="button"
      data-bs-toggle="collapse" data-bs-target="#navMenu">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navMenu">
      <ul class="navbar-nav ms-auto gap-2">
        <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#courses">Courses</a></li>
      </ul>
      <a href="#enroll" class="btn btn-success btn-sm ms-3">Get Started</a>
    </div>
  </div>
</nav>
```

The navbar collapses to a hamburger menu on mobile and expands on large screens (navbar-expand-lg).

Responsive Grid — Course Cards

Six course cards are laid out using Bootstrap's 12-column grid. Each card uses col-sm-6 col-lg-4 so they display one column on mobile, two on tablet, and three on desktop. The h-100 class makes all cards the same height; d-flex flex-column + mt-auto pins the price/button to the card bottom:

```
<div class="row g-4">
  <div class="col-sm-6 col-lg-4">
    <div class="card bg-black border-secondary h-100">
      <div class="card-body d-flex flex-column gap-2">
        <span class="badge bg-success bg-opacity-25 text-success align-self-start">25
Days</span>
        <h5 class="card-title fw-bold text-white">AI for Everyone</h5>
        <p class="card-text text-secondary small">Understand and apply AI in real
business contexts.</p>
        <div class="mt-auto d-flex justify-content-between align-items-center">
          <span class="text-success fw-bold">NPR 15,000</span>
          <button class="btn btn-outline-success btn-sm">Enroll</button>
        </div>
      </div>
    </div>
  </div>
  <!-- 5 more identical col blocks for other courses -->
</div>
```

Bootstrap Breakpoint	Screen Width	Cards per row in demo
Default (col-12 implied)	< 576px	1 card per row (stacked)
col-sm-6	≥ 576px	2 cards per row
col-lg-4	≥ 992px	3 cards per row

Hero, Stats, Form and Alert

Stats Row

```
<!-- Four equal-width stat boxes, stacked on mobile, side-by-side on md+ -->
<div class="row justify-content-center g-3 mt-5">
  <div class="col-6 col-md-3">
    <div class="p-3 rounded-3 border border-secondary bg-black">
      <div class="fs-3 fw-bold text-success">10,000+</div>
      <div class="small text-secondary">Learners</div>
    </div>
  </div>
  <!-- Repeat for 50+ Courses, 95% Placement, 120+ Reviews -->
</div>
```

Enrolment Form

```
<div class="card bg-black border-secondary p-4">
  <div class="row g-3">
    <div class="col-sm-6">
      <label class="form-label text-secondary small">First Name</label>
      <input type="text" class="form-control bg-dark text-white border-secondary" />
    </div>
    <div class="col-12">
      <label class="form-label text-secondary small">Select Course</label>
      <select class="form-select bg-dark text-white border-secondary">
        <option>AI for Everyone</option>
        <option>Cybersecurity</option>
      </select>
    </div>
    <div class="col-12 d-grid">
      <button class="btn btn-success btn-lg"
        onclick="this.textContent='✓ Submitted!';this.disabled=true;">
        Send Message
      </button>
    </div>
  </div>
</div>
```

Alert Banner

```
<div class="alert alert-success d-flex align-items-center
  justify-content-between flex-wrap gap-3" role="alert">
  <div>
    <strong>Free Career Counseling!</strong>
    Enroll today and get a free session with our industry experts.
  </div>
  <a href="#enroll" class="btn btn-success btn-sm">Claim Now</a>
</div>
```

Bootstrap utility Classes Used

Utility Class	What it Does	Used In
bg-dark / bg-black	Dark background colour	Cards, inputs, navbar, page body
text-success / text-secondary	Green or grey text	Headings, labels, badge text
d-flex / d-grid	Display flex or grid	Card footer layout, full-width button
justify-content-between	Space items apart on main axis	Price + button row, navbar, alert
align-items-center	Centre items on cross axis	Navbar, alert, card footer
ms-auto / mt-auto	Push element to far end	Nav links right; button to card bottom
gap-2 / gap-3 / g-4	Spacing between flex/grid children	Nav items, row gutters
fw-bold / small / fs-3	Font weight and size utilities	Card titles, labels, stat numbers
rounded-3 / border-secondary	Border radius and colour	Stat boxes, card borders
px-4 / py-5 / p-3	Horizontal/vertical/all padding	Buttons, sections, stat boxes

Summary

The **CodeCraft Bootstrap demo** is a recreation of the training website built entirely with **Bootstrap 5**, without using any custom CSS. The project highlights the effective use of Bootstrap components and utilities to create a responsive and modern web interface.

Key features included in the project:

- Bootstrap CSS and JavaScript integrated through the CDN.
- A responsive navigation bar with a collapsible menu and mobile-friendly hamburger icon.
- A flexible 12-column grid system using col-sm-6 and col-lg-4, allowing content to adapt from one column on small screens to three columns on large devices.
- Bootstrap card components containing badges, headings, descriptive, text, and equal-height layouts using utility classes such as h-100 and mt-auto.
- Styled form elements including text inputs, labels, and dropdown menus created with from-control, from-label, and from-select.
- A success alert section using the alert-success class combined with bootstrap's flex utilities for better alignment.
- Extensive use of bootstrap utility classes for spacing, typography, colors, and layout, helping maintain a clean and consistent design throughout the website.

Overall, the project demonstrates a solid understanding of Bootstrap 5 and its ability to build a fully responsive website without the need for additional CSS styling.